

★ WEEK 9 · PART 3 of 3

The Kalman Filter in Practice

Two Flagship Applications: Pairs Trading and Nowcasting

A self-adjusting hedge ratio · a real-time read on the economy

With the recursion and the gain in hand, we put the filter to work on two landmark problems: an adaptive hedge ratio for pairs trading, and nowcasting — estimating the economy's state right now from a stream of noisy, mixed-frequency data.

3

PART 3 · IN PRACTICE

Two Faces of One Filter

A dynamic hedge ratio that keeps a pairs trade market-neutral, and nowcasting that reads current GDP in real time from a ragged stream of indicators — both the same predict-update engine.

From the Recursion to Real Use

Parts 1-2 built it; Part 3 applies it twice

1

We have the full filter

Predict, update, the gain, and the covariance. The machinery is complete — now we use it on two very different problems.

2

Example 1 — pairs trading

Let the hedge ratio itself drift as a hidden state, so the spread stays market-neutral as the relationship evolves.

3

Example 2 — nowcasting

Read the current state of the economy in real time from many noisy, mixed-frequency indicators that arrive on a ragged schedule.

Part 3 in one line: the same recursion powers an adaptive hedge and a real-time economic nowcast.

Two Applications, One Filter

Different fields, identical machinery

Pairs trading and nowcasting look unrelated — one is a trading strategy, the other a macro forecast. Yet both treat what they want as a hidden state seen through noisy data, and both use the very same Kalman recursion.

Example 1 — pairs trading

- The hidden state is a drifting hedge ratio.
- The observations are the two asset prices.
- Goal: keep the traded spread market-neutral.

Example 2 — nowcasting

- The hidden state is the current state of the economy.
- The observations are many mixed-frequency indicators.
- Goal: estimate current-quarter GDP, right now.

The Shared Capability: Denoising

Both extract a clean state from noise

What the filter always does

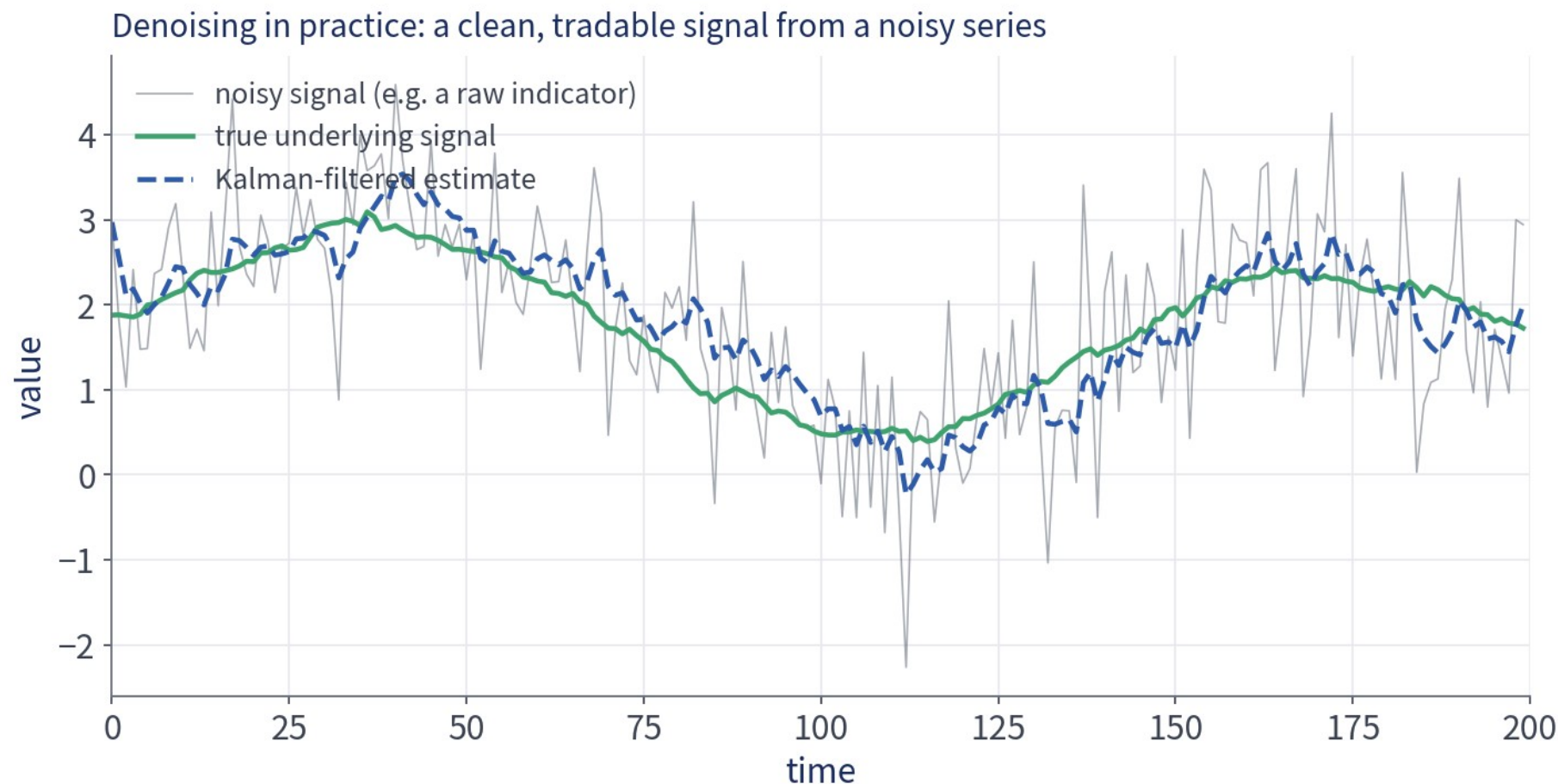
- Treats the quantity you want as a hidden state.
- Fuses a model with noisy data, weighted by uncertainty.
- Returns a clean, real-time estimate with error bars.

Why this generalizes

- A hedge ratio, a factor, a level — all just hidden states.
- The observation model changes; the recursion does not.
- Learn it once, apply it across finance and macro.

The Core Move: Signal From Noise

In both applications the filter does the same thing — a raw, noisy series (grey) becomes a clean estimate of the underlying hidden state (dashed), tracking the truth (green) in real time.



A Regression That Adapts

Letting the hedge ratio drift over time

Ordinary regression fixes a coefficient for all time. But the relationship between two assets drifts. The Kalman filter lets the hedge ratio itself be a hidden state that evolves — a regression that quietly updates.

The shift in view

- The hedge ratio is no longer one fixed number.
- It becomes a state with its own gentle dynamics.
- The filter re-estimates it at every observation.

Why it matters

- Market betas and exposures drift with the regime.
- A static estimate is stale the moment things move.
- A dynamic hedge ratio stays current, automatically.

The Dynamic Regression as State Space

Make the hedge ratio the state

dynamic regression: $Y_t = \beta_t X_t + \varepsilon_t$, $\beta_t = \beta_{t-1} + \eta_t$ (β is the state)

The two equations

- The observation: Y is beta-times- X plus noise.
- The state: beta follows a slow random walk.
- It is exactly a state space model — so Kalman applies.

The single knob

- How fast beta may drift is set by its process noise.
- Large: beta adapts quickly but jitters.
- Small: beta is stable but slow to move.

Why a Rolling Window Isn't Enough

The usual quick fix, and its flaws

The rolling-window fix

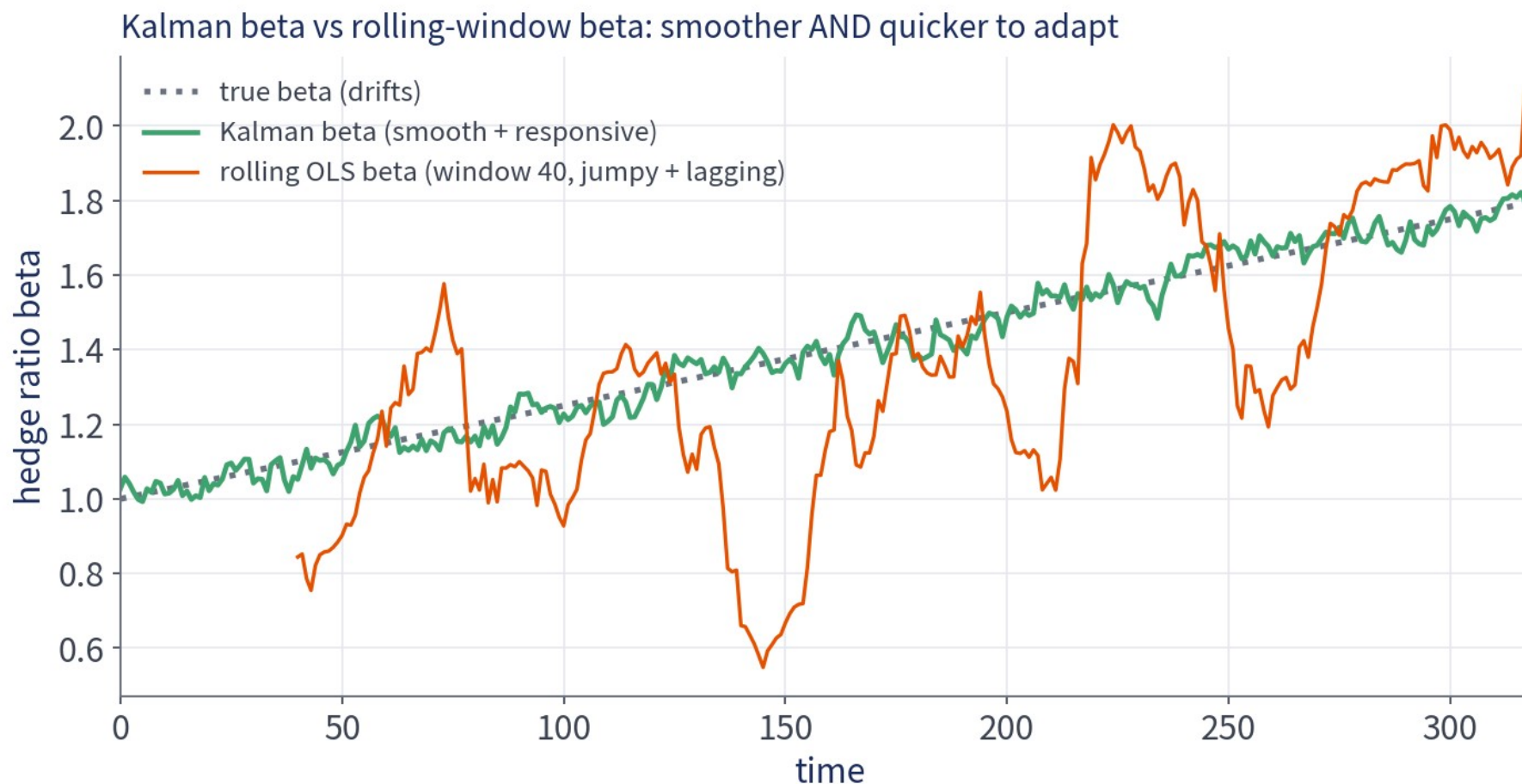
- Re-estimate the coefficient over the last N points.
- It does adapt — but only crudely.
- The window length is an arbitrary, awkward choice.

Why Kalman is better

- No window: it weights all history by relevance.
- It reacts faster AND stays smoother.
- It gives principled uncertainty, not just a point.

Kalman Beta vs Rolling-Window Beta

The true hedge ratio drifts (dotted). A rolling OLS beta (orange) is jumpy and lags; the Kalman beta (green) is both smoother and quicker to adapt — with no window length to guess.



Stage III: The Time-Varying Hedge Ratio

The thread gains a self-adjusting equilibrium

The pairs trade has had volatility-scaled bands (Stage I) and a cointegration anchor (Stage II). Stage III lets the hedge ratio itself drift — so the spread stays centered even as the relationship evolves.

The problem it solves

- A static hedge ratio slowly goes stale as beta drifts.
- The 'market-neutral' spread starts to leak exposure.
- Re-fitting by hand is clumsy and always behind.

The Stage III fix

- Model the hedge ratio as a Kalman state.
- It updates every day, keeping the spread centered.
- The trade stays neutral as the market changes.

The Adaptive Spread and Signal

A drifting hedge, a familiar trade

adaptive spread: $z_t = Y_t - \beta_t X_t \Rightarrow$ trade $\frac{z_t - \mu}{\sigma}$: enter $|\cdot| \geq 2$, exit ≈ 0

The spread

- Form the spread with the CURRENT filtered hedge ratio.
- Because beta adapts, the spread stays centered.
- It is a cleaner, more stable quantity to trade.

The signal

- Z-score the adaptive spread as before.
- Enter when it is stretched, exit as it reverts.
- The mechanics are Stage II's; the hedge is now live.

Why the Adaptive Spread Is Cleaner

A hedge that keeps up

Static hedge (Stage II)

- One beta fixed over the whole sample.
- As the true ratio drifts, the spread decenters.
- Signals degrade; false extremes appear.

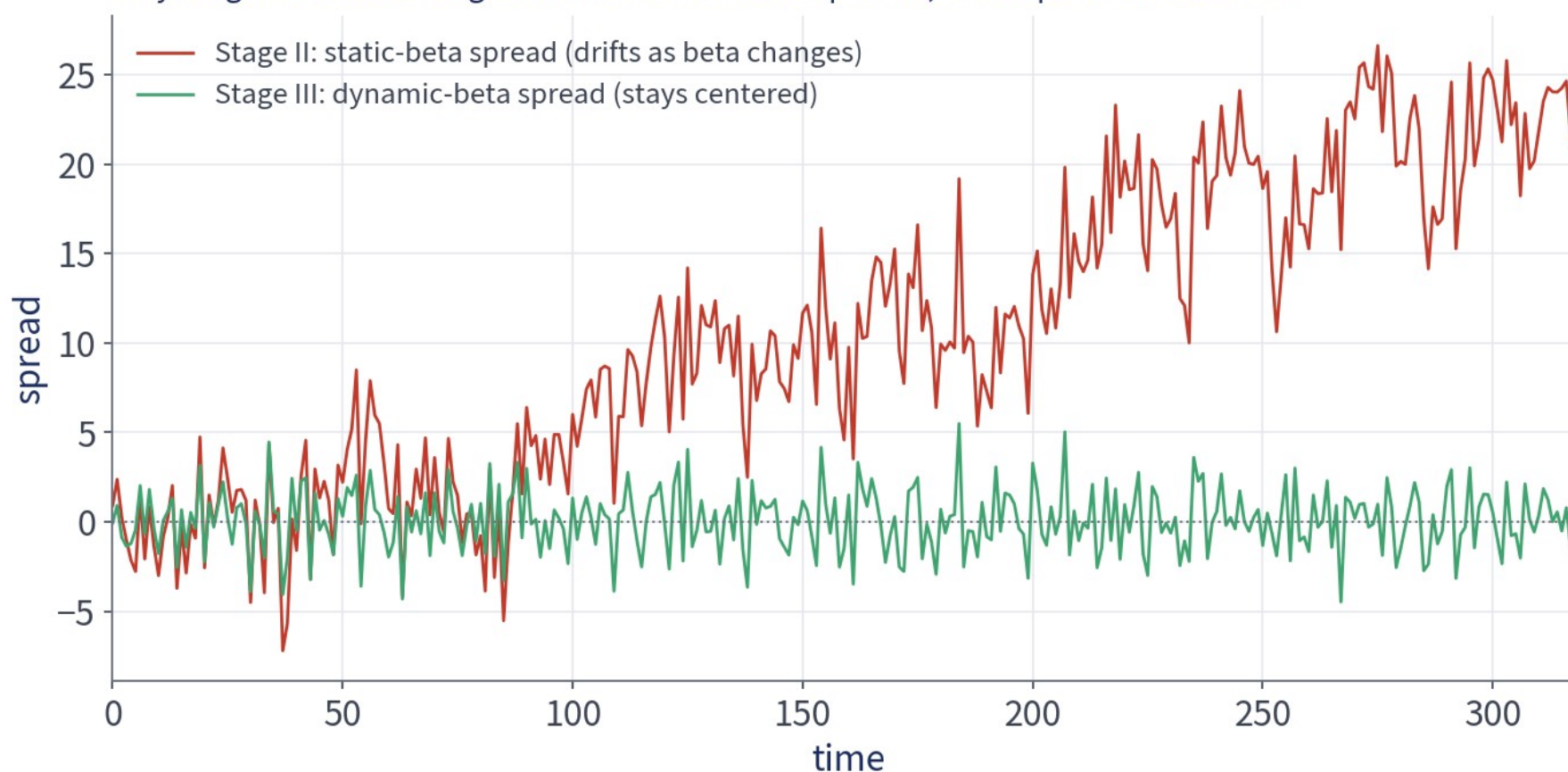
Dynamic hedge (Stage III)

- Beta tracks the true ratio in real time.
- The spread stays centered on zero.
- Signals stay meaningful as the regime shifts.

Static vs Adaptive Spread

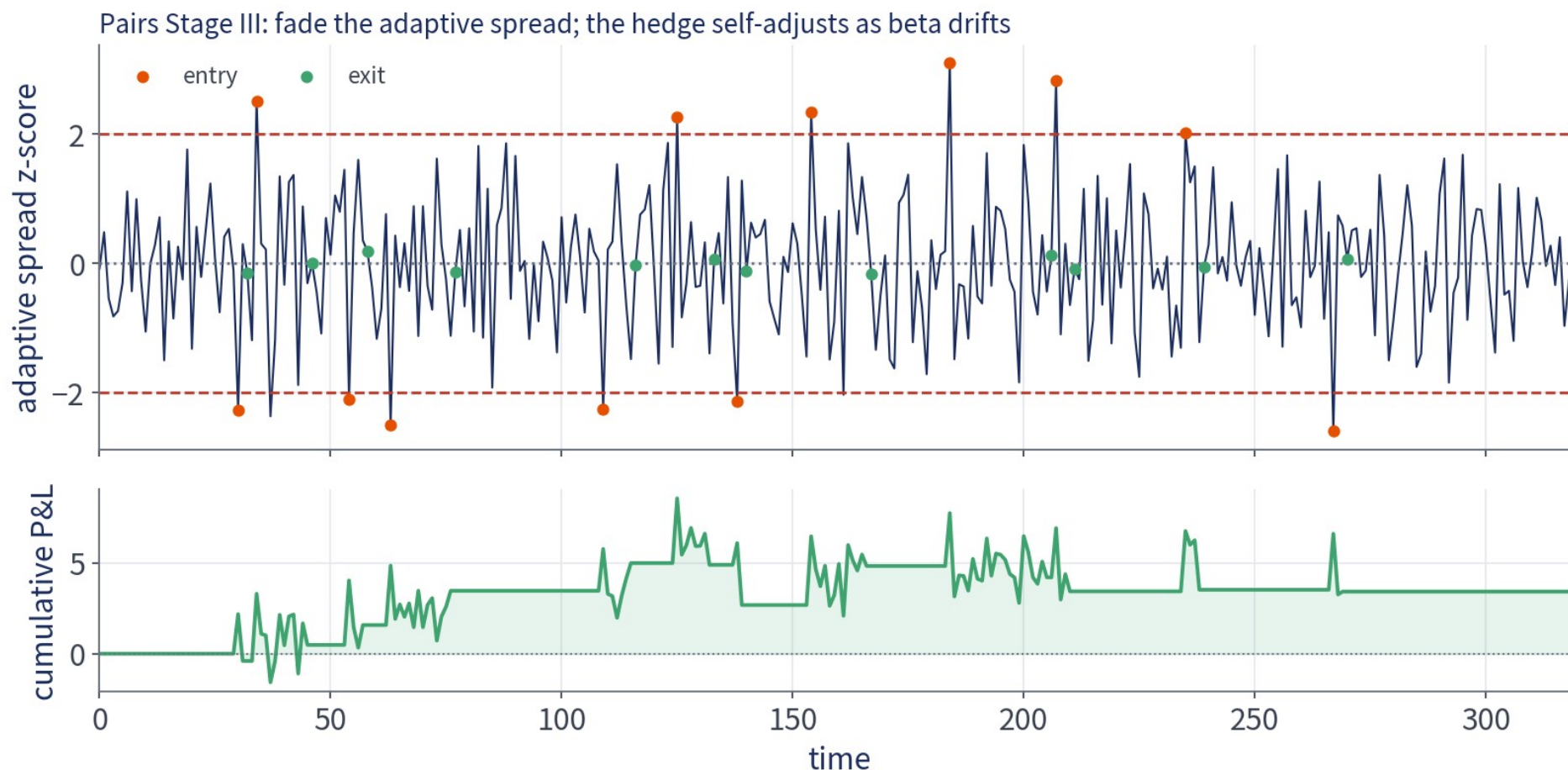
As the true hedge ratio drifts, the static-beta spread (red) leaks away from zero — producing false signals. The dynamic-beta spread (green) stays centered, keeping the trade honest.

Why Stage III: a static hedge leaks as the relationship drifts; an adaptive one does not



Pairs Stage III, in Action

Fade the adaptive spread's z-score at the bands and exit near zero. Because the hedge self-adjusts, the spread reverts cleanly and the cumulative P&L (below) accrues steadily.



Stage III vs Stage II

What the Kalman hedge adds

Aspect	Stage II (Week 7)	Stage III (Week 9)
Hedge ratio	static — one OLS beta	dynamic — a drifting state
Adapts to drift	no	yes, in real time
Spread when beta shifts	leaks off-center	stays centered
Underlying tool	cointegration + VECM	the Kalman filter

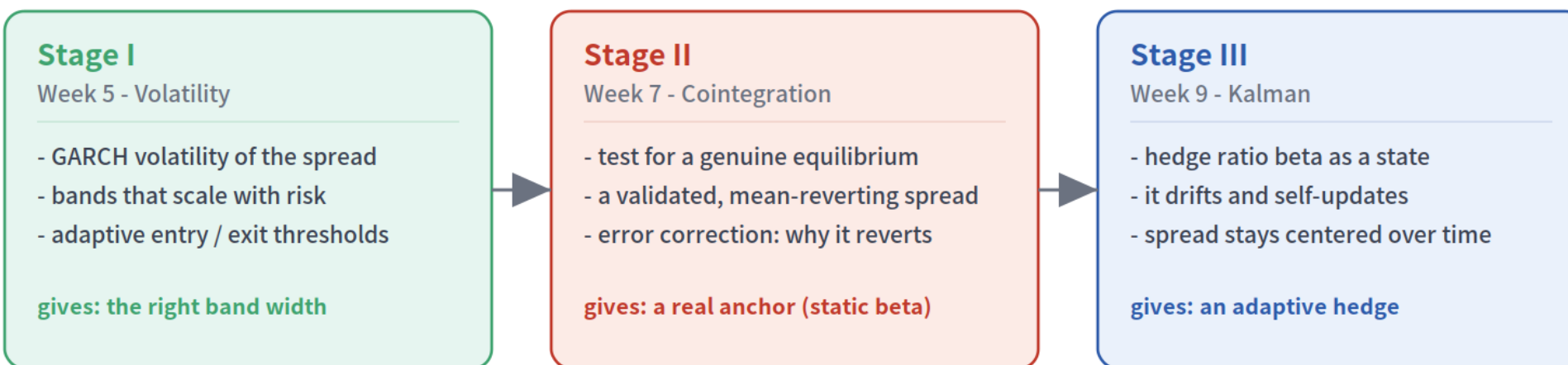
Stage III does not replace the cointegration anchor — it lets that equilibrium itself evolve.

The Complete Pairs Journey

Three stages across three weeks: volatility-scaled bands (W5), a cointegration anchor (W7), and a Kalman-adaptive hedge (W9). Each layer deepens the same market-neutral trade.

The Complete Pairs Journey

Three weeks, three stages - each adds a layer to the same trade.



The same idea, deepened each week

Find a pair, size the bands by volatility, anchor it with cointegration, and let the hedge ratio adapt.

Stage III does not replace II - it lets the equilibrium itself evolve, keeping the trade market-neutral.

What Is Nowcasting?

A real-time read on the economy

Official GDP arrives with a long lag and big revisions. Nowcasting estimates it RIGHT NOW — the present, not the future — by fusing the many timelier indicators that trickle out during the quarter. It is a landmark Kalman-filter application.

The idea

- Predict the present, before the official number exists.
- Use employment, surveys, production, sales — as they release.
- Introduced by Giannone, Reichlin and Small (2008).

Who uses it

- Central banks run live nowcasts of GDP.
- The New York and Atlanta Fed publish such models.
- It is now standard practice in policy and markets.

Mixed Frequency and the Ragged Edge

Why nowcasting is hard

Mixed frequency

- GDP is quarterly; most indicators are monthly or weekly.
- They must be combined despite different clocks.
- You need a model that speaks all frequencies at once.

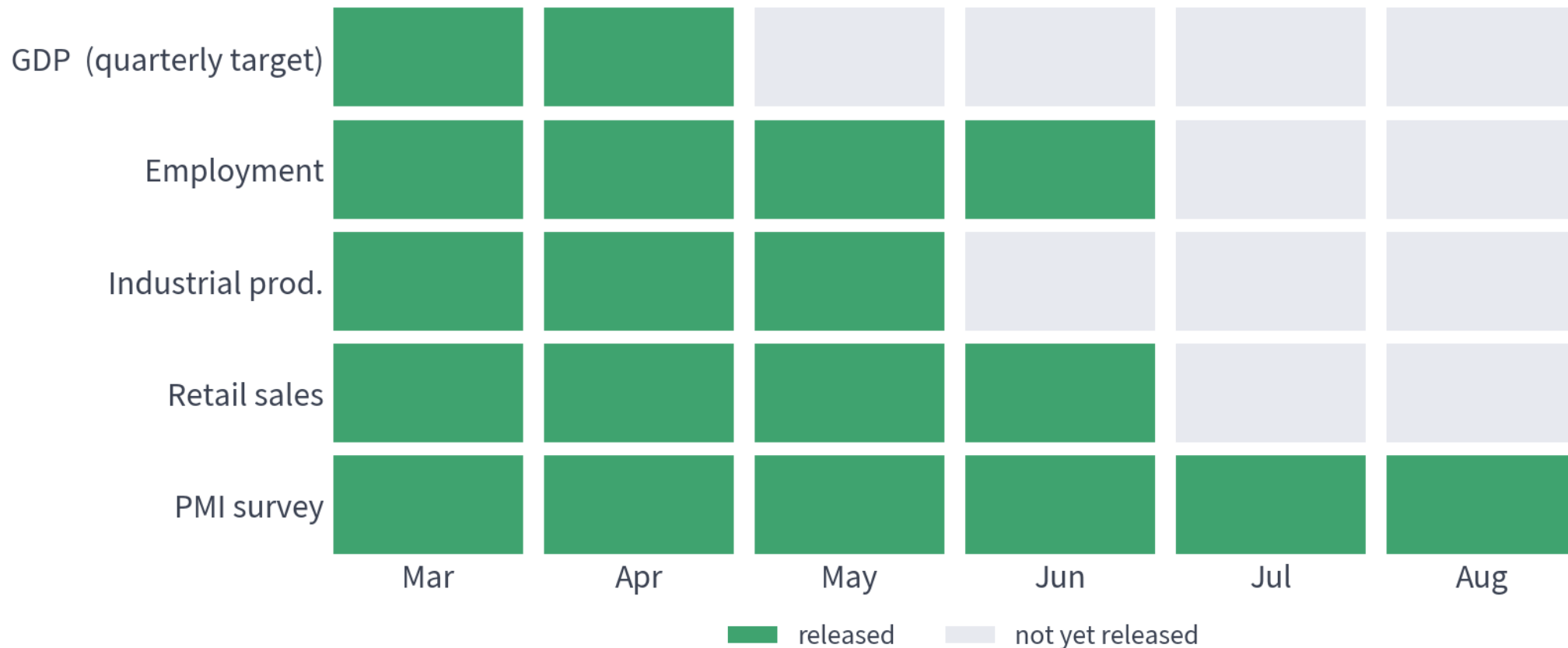
The ragged edge

- Indicators release on different schedules and delays.
- At any moment, each series ends at a different date.
- So the data panel has a jagged, incomplete edge.

The Ragged Edge

At any moment, series end at different dates: surveys are timely, hard data lag, GDP lags most. This jagged pattern of missing recent values is exactly what the Kalman filter handles gracefully.

The ragged edge: at any moment, series have different last-available dates



Why the Kalman Filter Is Natural Here

Missing data and real-time updating, built in

Two properties make the filter perfect for nowcasting: it handles missing data by simply skipping the update, and it updates in real time as each release arrives. The ragged edge that breaks other methods is a non-issue.

Missing data, handled

- If a series hasn't been released, skip its update step.
- The state still evolves; you just don't correct it there.
- The ragged edge needs no special patching.

Real-time, by design

- Each new release triggers one update — no full refit.
- The nowcast revises the instant data lands.
- Exactly the recursive advantage from Part 1.

The Dynamic Factor Model

One common factor drives many indicators

$$y_{i,t} = \lambda_i f_t + \varepsilon_{i,t}$$

each indicator loads on a common factor

$$f_t = A_1 f_{t-1} + \dots + \eta_t$$

the factor (state) evolves

The structure

- A few common factors capture the economy's state.
- Each indicator is that factor times a loading, plus noise.
- The factor evolves over time like a small VAR.

Why a factor

- Dozens of indicators share one underlying signal.
- The factor IS the hidden 'state of the economy'.
- Estimating it filters out each series' idiosyncratic noise.

Casting It in State Space

The factor is the state; the filter does the rest

ragged edge: if $y_{i,t}$ not yet released $\Rightarrow$ skip its update (treat as missing)

The mapping

- The common factor becomes the hidden state.
- Each indicator is an observation of it (with its loading).
- Mixed frequencies fit as observations at different steps.

Ragged edge, dissolved

- A not-yet-released value is simply a missing observation.
- The filter skips its update and moves on.
- No imputation, no truncation — it just works.

The Nowcast, Defined

The best estimate of the present

$$\text{nowcast} = \mathbb{E}[\text{GDP}_{\text{current}} \mid \text{all data released so far}]$$

What it is

- The expected current-quarter GDP given all data so far.
- It is a filtered estimate — causal, using only released data.
- It comes with an uncertainty band, like any filter output.

How it behaves

- Early in the quarter it is uncertain — little data yet.
- It sharpens as more indicators release.
- It converges toward the eventual official print.

The Nowcast Updates in Real Time

A living estimate, revised on every release

The nowcast is not a one-shot forecast — it is a running estimate that revises each time a new figure is published. As the quarter fills in, the point estimate settles and its uncertainty band narrows.

Through the quarter

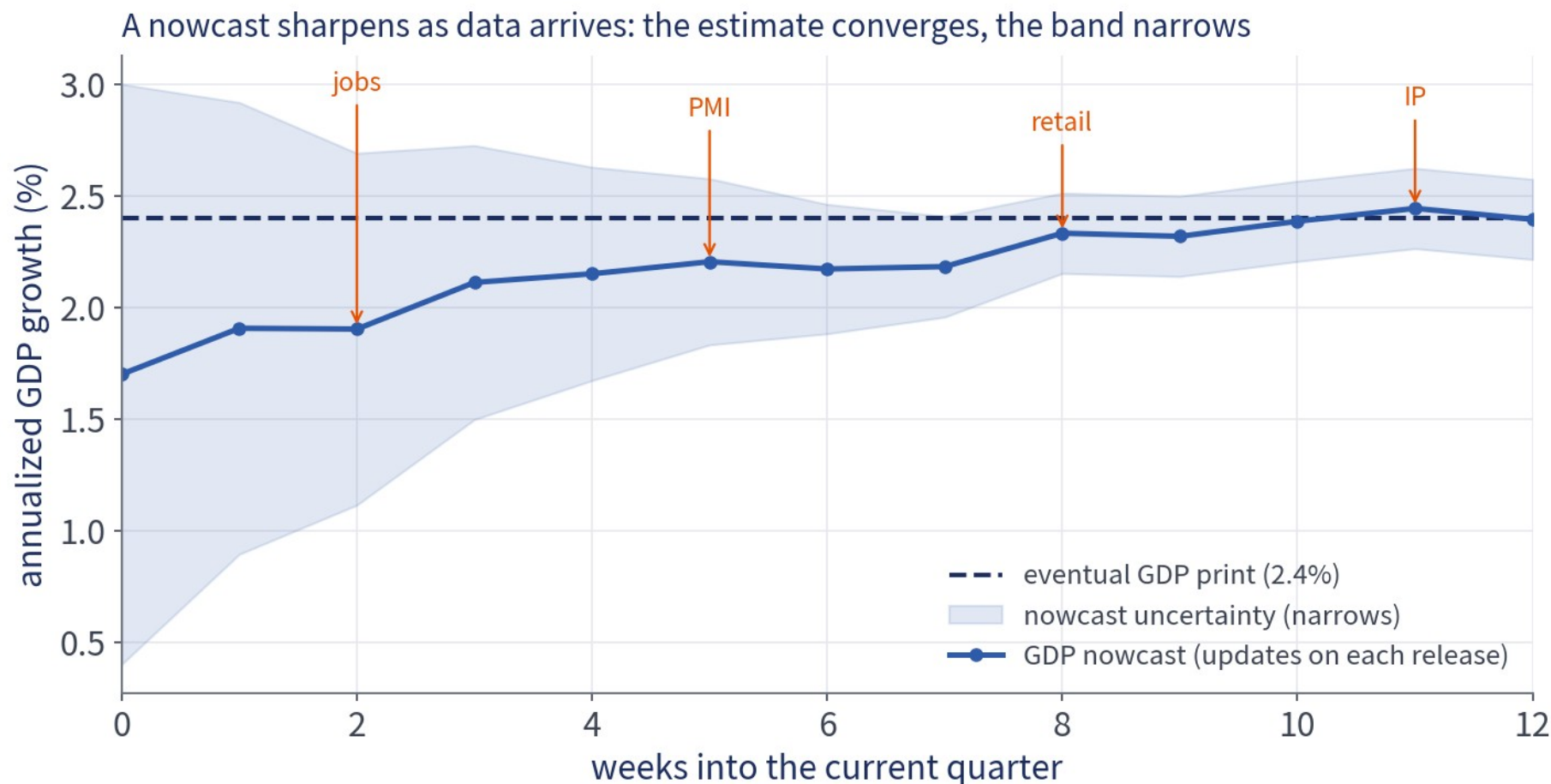
- Start from a prior with a wide uncertainty band.
- Each release nudges the estimate and tightens the band.
- By quarter's end it is close to the true value.

What this gives policymakers

- A continuously current read on growth.
- An early warning when momentum shifts.
- A principled way to weigh conflicting new data.

A Nowcast Sharpening Through the Quarter

Each data release (jobs, PMI, retail, industrial production) updates the nowcast. The point estimate converges toward the eventual GDP print and the uncertainty band steadily narrows.



'The News': Each Release Is a Surprise

The nowcast revision is an innovation

$$\text{nowcast revision} = K_t \times \underbrace{(\text{release} - \text{its forecast})}_{\text{the 'news' = innovation } v_t}$$

The connection

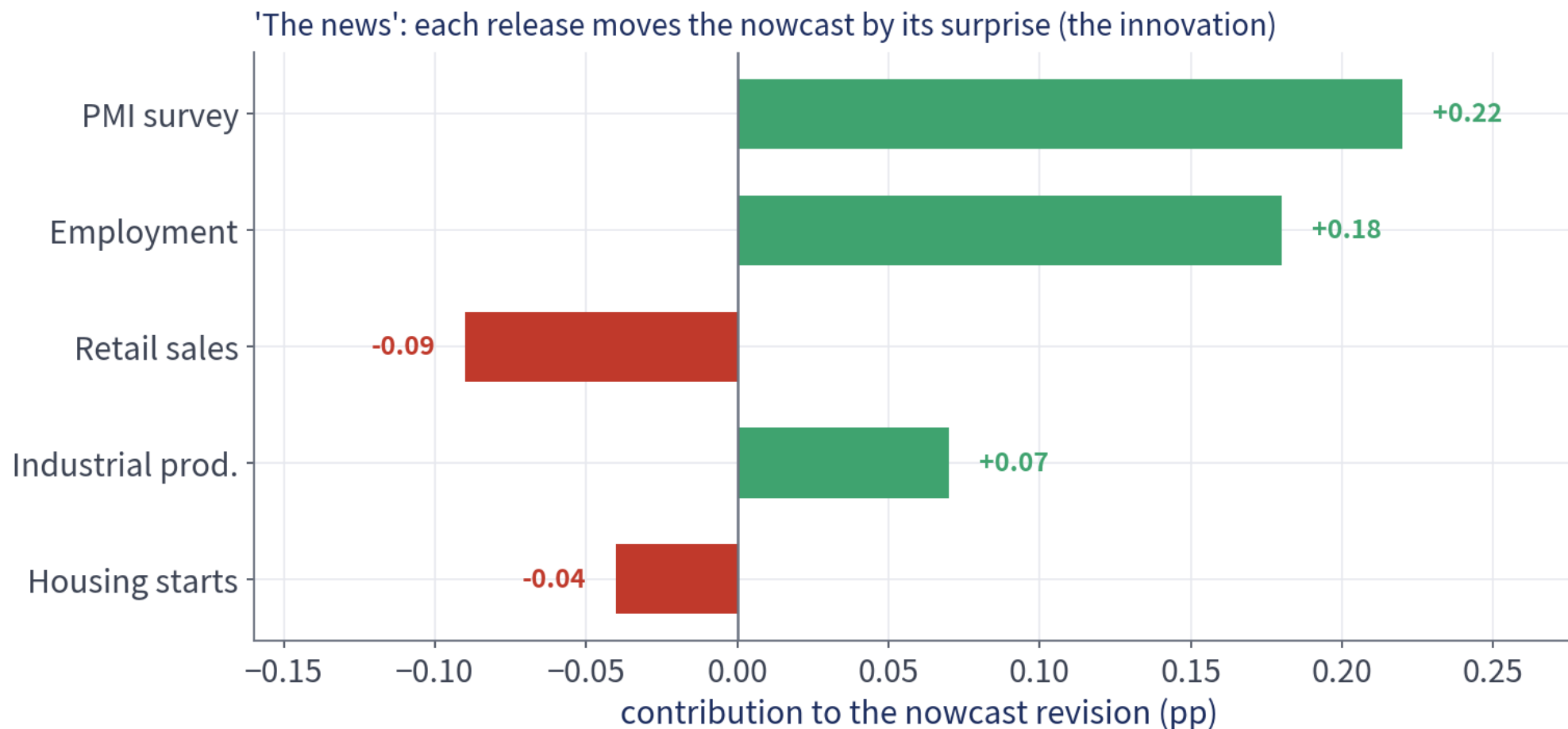
- A release matters only to the extent it SURPRISES.
- That surprise is exactly the Kalman innovation from Part 1.
- The revision is the gain times that surprise.

Why practitioners love it

- You can attribute each nowcast move to a specific release.
- An in-line number moves the nowcast little; a shock moves it a lot.
- This 'news' decomposition is a standard nowcasting output.

Decomposing the Nowcast Into News

Each recent release contributes to the nowcast revision by its own surprise — positive (green) or negative (red). The filter turns a stream of data into an interpretable set of contributions.



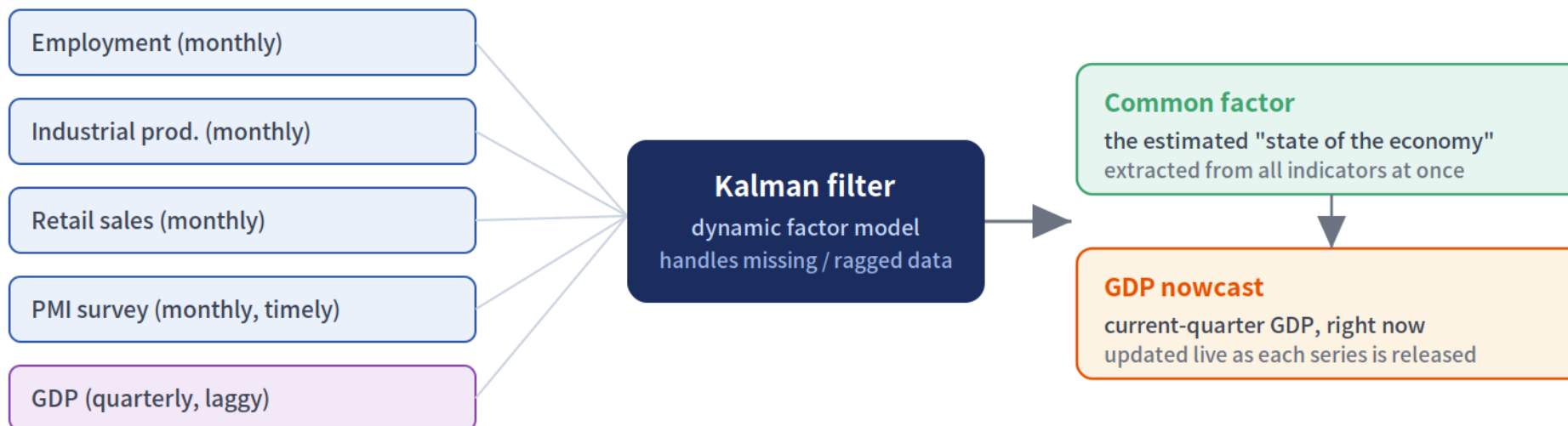
Nowcasting With the Kalman Filter

Many mixed-frequency, ragged-edge indicators feed one filter, which extracts the common factor — the state of the economy — and reads off a current-quarter GDP nowcast, updated on every release.

Nowcasting With the Kalman Filter

Many mixed-frequency, ragged-edge indicators feed one filter, which reads the state of the economy now.

Indicators (mixed frequency)



Each release is 'news'

Every new data point updates the nowcast by its surprise - the Kalman innovation. This is the same engine as Parts 1-2.

Nowcasting in the Real World

From a 2008 paper to central-bank dashboards

The literature

- Giannone, Reichlin and Small (2008) framed the method.
- Banbura, Giannone and Reichlin extended it to large panels.
- It rests on dynamic factor models estimated by the Kalman filter.

The practice

- Central banks publish live GDP nowcasts (e.g. Federal Reserve).
- The Atlanta Fed's GDPNow is a widely-watched example.
- Markets track these nowcasts around major data releases.

Pairs Trading vs Nowcasting

The same machinery, two problems

Aspect	Pairs trading (Ex. 1)	Nowcasting (Ex. 2)
Hidden state	a drifting hedge ratio	the state of the economy (a factor)
Observations	two asset prices	many mixed-frequency indicators
Key filter feature	tracking a drifting parameter	handling missing / ragged data
Output	a centered, tradable spread	a real-time GDP estimate

Different fields, one recursion: a hidden state, noisy observations, and predict-update.

A Word of Caution

The filter is only as good as its inputs

In both applications the filter is optimal ONLY if the model and noise assumptions hold. A wrong model or badly chosen noise levels will produce a confident but wrong answer — so always check the innovations and validate out of sample.

What to watch

- Innovations should look like white noise — no leftover structure.
- Choose the process and measurement noise with care.
- Validate on held-out data, not just in-sample fit.

When to reach elsewhere

- Strong nonlinearity → extended or unscented filters.
- Discrete, switching regimes → hidden Markov models (Week 10).
- Heavy-tailed shocks → robust or particle filters.

Build the Stage III Pairs Trade

A self-adjusting hedge ratio, end to end

GOAL Estimate a time-varying hedge ratio with a Kalman filter and back-test the adaptive pairs trade.

[INPUT] Prompt · Claude Code

Write and run `pairs_stage3.py` that:

1. models the hedge ratio β_t as a Kalman state (dynamic regression of Y on X);
2. forms the ADAPTIVE spread $Y - \beta_t X$ and compares it to the STATIC-beta spread;
3. z-scores the adaptive spread and back-tests: enter at $|z| \geq 2$, exit near 0, with a stop;
4. plots β_t , both spreads, and cumulative P&L for static vs dynamic.

Report which hedge keeps the spread centered.

How to work with it

- β_t should drift smoothly, not jitter.
- The adaptive spread should stay centered.
- Check innovations are white noise.
- This completes the pairs-trading thread.

Deliverable: `pairs_stage3.py` + a dynamic beta, both spreads, and a static-vs-dynamic back-test.

Build a Simple GDP Nowcaster

A dynamic factor model with ragged data

GOAL Fit a small dynamic factor model by Kalman filter and produce a nowcast that updates as data arrives.

[INPUT] Prompt · Claude Code

Write and run `nowcast_demo.py` that:

1. builds a few monthly indicators driven by one common factor, plus a quarterly GDP target;
2. fits a dynamic factor model in state-space form (`statsmodels DynamicFactor` or a hand-coded Kalman filter) allowing MISSING values;
3. introduces a ragged edge (drop the most recent values of the laggier series);
4. produces the nowcast and its band, and shows it update as each 'release' is added back.

Report how the nowcast and band change per release.

How to work with it

- Missing values must be allowed in the filter.
- The band should narrow as data fills in.
- Each release should move the nowcast by its surprise.
- This is nowcasting in miniature.

Deliverable: `nowcast_demo.py` + a nowcast that updates and tightens as releases arrive.

Three Questions Before You Finish

If you can answer these, Week 9 has landed

Q1	In Stage III, what is treated as the hidden state?	The drifting hedge ratio
Q2	How does the Kalman filter handle a not-yet-released indicator?	Skip its update (missing)
Q3	A nowcast jumps on a data release. What moved it?	The surprise (innovation)

All three come from the dynamic hedge, ragged-edge handling, and the news decomposition.

Week 9 in Three Ideas

Pulling the whole week together

1

Predict, update, and the gain

Parts 1-2: the two-step recursion, the innovation that drives it, and the Kalman gain that optimally weighs model against data — the whole filter in five equations.

2

One recursion, two flagship applications

Part 3: a drifting hedge ratio for pairs trading (Stage III), and a real-time GDP nowcast from ragged, mixed-frequency data. Both treat the target as a hidden state.

3

The same ideas, everywhere

Missing data is just a skipped update; a data 'surprise' is an innovation; a drifting parameter is a state. Master the recursion and these applications follow.

Next — Week 10: Hidden Markov Models, for a hidden state that switches between discrete regimes.